

SUDARA T S M

✉ sudharshinisudara@gmail.com | 🌐 Sudara17 | in sudara17 | 📞 +91-9345776397

Executive Summary

AI & Data Science student with hands-on experience building LLM-powered applications, multimodal emotion recognition systems, and real-time AI assistants. Skilled in prompt engineering, NLP, RAG pipelines, and deploying applications using Python and Flask. Seeking opportunities to apply technical expertise toward impactful AI-driven solutions.

Experience

Connected Value Health Solutions Pvt. Ltd., Coimbatore Jun 2025 – Jul 2025
Gen AI Intern

- Built resume parsing and semantic extraction modules using **LLM embeddings**, validated with **10+ real-world resume samples**.
- Designed **RAG-powered chatbot workflows** enabling contextual responses and structured multi-step resume guidance.
- Optimized backend inference by reducing response latency by **150ms** through prompt refinement and efficient token usage.
- Developed reusable **LangChain integration modules** to enhance workflow automation and maintainability.

Acmegrade Jun 2024 – Aug 2024
Web Development Intern

- Developed responsive UI screens and interactive components using **HTML, CSS, JavaScript** aligned with modern UX principles.
- Improved page performance and stability through iterative **load optimization and cross-browser compatibility** testing.

Education

B.S. Abdur Rahman Crescent Institute of Science and Technology 2022 – 2026
B.Tech in Artificial Intelligence and Data Science

Projects

Emotion-Aware Assistant

Tech stack: Python, TensorFlow, Flask, Whisper, gTTS, OpenCV, MediaPipe, PyTorch, NLTK/VADER

- Built a multimodal AI assistant capable of detecting emotions from **voice, text, and facial expressions**.
- Developed speech emotion recognition using **MFCC features and TensorFlow models**, enabling classification of 6 emotional states.
- Integrated **Whisper ASR and transformer-based sentiment analysis** for accurate text-based emotion inference.
- Implemented **OpenCV + MediaPipe facial analysis** pipeline for real-time expression detection.
- Achieved **89% multimodal accuracy** in real-time emotion detection across speech, text, and facial cues for accessibility-focused interaction.

Waste Classification using CNN

Tech stack: *Python, Keras, TensorFlow*

- Developed a **Convolutional Neural Network (CNN)** model in **Keras** for classification of biodegradable vs non-biodegradable waste using augmented image datasets.
- Evaluated model achieving **92.5% test accuracy** with cross-validation on sample waste image datasets.

AI Resume Assistant

Tech stack: *Python, LangChain, Groq LLM, FAISS, Streamlit*

- Built an AI-powered Resume Assistant using **LangChain** and **Groq LLM** to parse resumes and provide tailored content improvements.
- Integrated **ATS scoring**, **cover letter generation**, and **mock interview question generation** using NLP-based evaluation.
- Used **FAISS vector search** reducing retrieval time by **12%** while validating performance using **10+ resume test samples**.
- Deployed on **Render** with a lightweight inference backend for fast interaction.

Skills

AI/ML & NLP	Machine Learning, Deep Learning, NLP, LLMs (Groq, GPT), RAG, LangChain, FAISS, Transformers, Whisper, NLTK/VADER
Programming Languages	Python, SQL, JavaScript, HTML, CSS
Frameworks & Libraries	TensorFlow, Keras, PyTorch, Flask, Streamlit, Scikit-learn, Pandas, NumPy, Bootstrap, OpenCV, MediaPipe
Tools & Platforms	HuggingFace, PyMuPDF, Tesseract OCR, gTTS, Git, GitHub, Tableau, Power BI, AWS QuickSight
Deployment Platforms	Netlify, Vercel, Render, GitHub Pages

Certifications & Achievements

- **Research Publication (Springer Nature) & 3rd Place – ICISDIKSA 2026** 
Published and presented “AI-Powered Email Summarization and Assistant System using LLMs”, securing 3rd place in oral presentation
- **E-Business Certification** – NPTEL (IIT Kharagpur), 2025
- **Ignite India Program** – Wadhvani Foundation, 2025
- **Arduino Training** – IIT Bombay (Spoken Tutorial), 2025
- **Web Development Concepts** – Infosys Springboard, 2024
- **C Programming Training** – IIT Bombay (Spoken Tutorial), 2024